Effect of *In-situ* Soil Moisture Conservation Techniques, Soil Conditioner (Pusa Hydrogel) with Stress Management Practices on Growth and Yield of Rainfed Cotton

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ABSTRACT: Field experiments were conducted at Regional Research Station, Aruppukottai, Tamil Nadu Agricultural University, Coimbatore, Tamil Nadu during *rabi* season of 2016 and 2017 to study the impact of *in-situ* moisture conservation and stress management practices on soil moisture retention and productivity of cotton under rainfed Vertisol soil with the test variety SVPR - 2. The experiments were laid out in split plot design replicated thrice. The main plot treatments consisted of different *in-situ* moisture conservation measures *viz.,* Broad Bed and Furrows (I_1), Ridges and Furrows (I_2) and Compartmental Bunding (I_3). The subplot comprises the stress management practices *viz.,* soil application of pusa hydrogel @ 5 kg ha^{-1} (S_1), soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of 1% KCl (S_2), soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of 5% kaolin (S_3), soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} (S_4), soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of salicylic acid 100 ppm (S_5) and control (S_6). The results of this study showed that treatment combination of broad bed and furrow and soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} recorded significantly higher values of growth characters such as plant height, dry matter production, LAI and yield attributes *viz*., sympodial branches per plant, number of bolls plant^{-1}, boll weight and seed cotton yield 1,580 kg ha^{-1} (2016) and 1,943 kg ha^{-1} (2017).

Key words: Broad bed and furrows, pusa hydrogel, rainfed cotton

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Introduction

Cotton as a crop as well as a commodity plays an important role in the agrarian and industrial activities of the nation and has a unique place in the economy of our country. Cotton, popularly known as "White Gold" is cultivated mainly for fibre and additionally as an important source of edible oil. India devotes more area to cotton than any other country in the world. At present, India ranks first in area with 11.88 M ha, accounting 30% of world average and 22% (351 lakh bales of lint) of the world cotton production with a lint productivity of 568 kg ha^{-1} (DCD, 2017). Nearly 65% of the cotton crop is cultivated under rain fed condition in the country. In Tamil Nadu, 1.33 lakh ha area is under cotton cultivation with a production of 6.5 lakh bales with and lint productivity of 620 kg ha^{-1}. India has been the traditional home of cotton and their textiles. India has progressed substantially in improving both production and productivity of cotton over the last five years, transforming from a net importer of cotton, to the largest exporters, shipping 6.9 million bales (2015-16) followed by USA (Kannan *et al.*, 2017).

To combat adverse soil moisture scarcity conditions, matching integrated drought management practices need to be evolved. Water stress is one of the most important factor limiting crop productivity and adversely affecting square and boll formation, lint yield and fibre quality of cotton. The loss of productivity due to the erratic distribution of rainfall and insufficient irrigation has threatened cotton farming. Rainfed areas can be made productive and profitable by adopting improved technologies for rainwater conservation and harvesting and commensurate agricultural production technologies. Soil management practices are tailored to store and conserve as much rainfall as possible by reducing runoff and increasing storage capacity in the soil profile. The most efficient and cheapest way of conserving rainfall is to hold it *in-situ.* The principle behind the different *in-situ* moisture conservation practices is to increase the infiltration by reducing the rate of runoff, temporarily impounding the water on the surface of the soil to increase the opportunity time for infiltration and modifying the land configuration for inter plot water harvesting (Muthamilselvan *et al.*, 2006).

The moisture stress during crop growth period is the primary cause for the yield reduction in cotton. To improve the soil moisture availability, by reducing the evaporation losses and retaining the moisture in effective rooting zone. The soil application of super absorbent polymers (SAPs) is found to be the promising technology for rainfed areas. However, very limited research work has experimented on the use of polymers as soil moisture retainers. One of such developed product is 'Pusa hydrogel' which is first successful indigenous semi-synthetic super absorbent technology for conserving water and enhancing crop productivity and thereby increases the water use efficiency (IARI, 2012). To reduce transpiration losses, foliar application of nutrient formulations, growth regulators, anti transpirants *etc.,* are being

tried by many researchers. Keeping this in view, an attempt was made to study the impact of *in-situ* moisture conservation, stress management practices on crop growth indices and productivity of cotton under rainfed agroecosystem.

Materials and Methods

Field experiments were conducted at Regional Research Station, Aruppukottai, Tamil Nadu Agricultural University, Tamil Nadu during *rabi* season of 2016 and 2017 with the test variety SVPR-2. The experimental site comes under the southern agro-climatic zone of Tamil Nadu and geographically situated at 9º 33'N latitude and 78º 05' E longitude at an latitude of 50 m above mean sea level. North east monsoon season is found to be more favourable in Aruppukottai region since 42% of annual rainfall is being received during this monsoon season. The soil of the experimental fields was medium deep, well drained vertisol (Type Chromusterts) in texture. The soil low in available nitrogen, low in available phosphorus and high available potassium. All package of practices were carried out as per recommendation of (CPG, 2012).

The experiment was laid out in split plot design, replicated thrice. The main plot treatments consisted of different *in-situ* moisture conservation measures *viz.,* Broad Bed and Furrows (I_1), Ridges and Furrows (I_2) and Compartmental Bunding (I_3). The subplot comprises with stress management practices *viz.,* soil application of pusa hydrogel @ 5 kg ha^{-1} (S_1), soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of 1% KCl (S_2), soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of 5% kaolin (S_3), soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} (S_4), soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of salicylic acid 100 ppm (S_5) and control (S_6).

The data analysis for the probability occurrence of 30 years rainfall in a standard week showed that there is a possibility occurrence of getting more than 80% of consecutive dry spell is laid with 45th and 50th standard weeks. So, to avoid stress, foliar spray has given at these standard weeks during 2016 based on historical rainfall probability analysis by Markov Chain method. Biometric observations recorded on growth attributes *viz*., plant height, dry matter production and yield attributes *viz*., sympodial branches plant^{-1}, bolls plant^{-1}, boll weight and seed cotton yield, dry matter efficiency and harvest index.

The seed cotton yield was obtained from net plot area was shade dried, weighed at each picking and yields of all picking were added and calculated as kg per plot and then expressed in kilogram per hectare. The data obtained were subjected to statistical analysis and were tested at five per cent level of significance to interpret the treatment differences as suggested by Gomez *et al.* (1984).

Results and Discussion

Effect of *in-situ* moisture conservation measures and stress management practices on growth of cotton plant height

Among the different land configurations, plant height was higher under broad bed furrows followed by ridges and furrows and compartmental bunding. Increased plant height at BBF system was due to adequate availability of soil moisture especially around the root zone, which reduced soil moisture stress and enhanced normal physiological activities of the plant by maintaining better plant water status and increased nutrient uptake, which ultimately resulted in better plant growth. The similar kind of results were reported by Jat *et al.* (2012) and Vivek and Bosu (2014). Stress management practices also had a remarkable influence on plant height at all the crop growth stages during both the year. At early growth stages of cotton, application of soil conditioner registered significantly higher plant height than the control. In addition the foliar spray of PPFM @ 500 ml ha^{-1} recorded taller plants (114.4 cm in 2016 and 142.5 cm in 2017) (Table 1). The increased plant height was due to better retention of moisture and indirectly the availability

Table 1 : Effect of *in-situ* moisture conservation and stress management practices on plant height (cm) of rainfed cotton during *rabi* 2016 and 2017

Treatment	*Rabi* 2016				*Rabi* 2017			
	I_1	I_2	I_3	Mean	I_1	I_2	I_3	Mean
S_1	103.5	97.4	90.3	97.1	125.6	119.2	112.8	119.2
S_2	117.1	111.1	104.1	110.8	146.1	136.3	126.5	136.3
S_3	113.3	106.3	99.3	106.3	139.5	132.5	125.6	132.5
S_4	125.5	115.1	102.6	114.4	155.6	141.5	130.4	142.5
S_5	106.1	101.6	96.1	101.3	132.5	126.8	121.1	126.8
S_6	93.8	91.7	83.6	89.7	114.4	111.2	106.0	110.5
Mean	109.9	103.9	96.0		135.1	127.9	120.4	
	I	**S**	**I at S**	**S at I**	**I**	**S**	**I at S**	**S at I**
S.Ed	2.7	1.9	4.0	3.2	2.8	2.3	4.1	3.3
CD (P < 0.05)	5.7	3.4	12.2	6.6	6.4	4.1	12.7	6.7

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Table 2 : Effect of *in-situ* moisture conservation and stress management practices on LAI of rainfed cotton during *rabi* 2016 and 2017

Treatment	*Rabi* 2016				*Rabi* 2017			
	I_1	I_2	I_3	Mean	I_1	I_2	I_3	Mean
S_1	4.51	4.19	3.88	4.19	4.67	4.37	4.08	4.37
S_2	5.16	4.73	4.29	4.73	5.29	5.00	4.71	5.00
S_3	4.88	4.46	4.04	4.46	5.01	4.66	4.33	4.67
S_4	5.70	5.09	4.47	5.09	6.21	5.54	4.87	5.54
S_5	4.75	4.38	4.01	4.38	4.85	4.59	4.31	4.58
S_6	4.17	3.95	3.73	3.95	4.46	4.13	3.79	4.13
Mean	4.86	4.47	4.07		5.08	4.72	4.35	
	I	**S**	**I at S**	**S at I**	**I**	**S**	**I at S**	**S at I**
S.Ed	0.12	0.09	0.16	0.12	0.12	0.14	0.24	0.23
CD (P < 0.05)	0.32	0.18	0.39	0.25	0.34	0.27	0.54	0.48

of nutrients provided by hydrophilic polymer, where it might have helped to increase the activity of cell division, expansion and elongation, ultimately leading to increased plant height. Similar results have been reported by Al-Harbi *et al.* (1999) in cucumber and Sivapalan (2001) in soybean. Further, PPFM spray had a constructive role in increasing plant height through the production of growth promoting hormones (Holland, 1997). The increased plant height due to PPFM was also confirmed by Suresh Reddy (2002). The combination of BBF with soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} recorded taller plants and this might be due to BBF and pusa hydrogel acted as moisture holders which gave good microclimatic environment in rhizosphere soil (Table 1).

Leaf area index

Leaf area index (LAI) determines the total photosynthetic area available to the plant and also the quantum of source that would be ultimately available for translocation to the sink. Rainfed cotton recorded higher LAI in 2017 than 2016 at all the crop growth stages. In the present investigation, leaf area index was significantly higher under BBF system (4.86 and 5.08) when compared with ridges and furrows and compartmental bunding during both years of experimentation. The size of leaf is determined by the number and size of the cell of which the leaf is built and is influenced by soil moisture, nutrients and light. The BBF landform management system essentially reduces the velocity of runoff water and consequently increases opportunity time for water to infiltrate into the soil (Ferrero *et al.,* 2005). Since LAI is directly dependent on the size of leaves, any soil moisture stress during the growth period reduces LAI of the plant. Similar findings were also quoted by Singh *et al.* (1997). BBF helps not only conserving water but also acts as drainage channel during moisture surplus season to avoid water stagnation. Better aeration and moisture maintained in soil under BBF helped for better growth and higher LAI. With respect to stress management practices, soil application of pusa hydrogel @ 5 kg ha^{-1} registered higher LAI at 30 and 60 DAS. In later stages, soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} recorded significantly higher LAI of 4.41, 5.09 (2016) and 4.59, 5.54 (2017) at 90 and 120 DAS, respectively. As the water content of the plant decreases, cell shrinks and turgor pressure increases reduced cell volume, subsequently concentrates solutes in cells. Super absorbent polymers (SAPs) increased the turgor pressure inside the cells by maintaining sufficient amount of water as per plant need and thus causing increase in leaf area and other related parameters reported by Al-Harbi *et al.* (1999) and Yazdani *et al.* (2007). In addition to that the foliar application of PPFM helped in more leaf expansion, plant height and higher biomass, which led the increased LAI in this treatment. This is in accordance with the findings of Madhaiyan *et al.* (2006b) and Abd El-Gawad *et al.* (2015) (Table 2).

Dry matter production

The amount of dry matter produced depends on the rate of photosynthesis which in turn depends on large and efficient assimilating area. A gradual increase in dry matter production of crop was highly attributed to vigorous growth of crop in terms of gain in plant height and higher number of functional leaves per plant (Memon *et al.,* 2007). At all stages of crop growth, the increased plant height and higher LAI paved way for production photosynthate under BBF followed by ridges and furrows and compartmental bunding during both the years of investigation. BBF accounted for higher dry matter production (4698 in 2016 and 5492 kg ha^{-1} in 2017) as compared to other moisture conservation methods. This may be possibly due to better soil moisture availability to the crop at all growth stages. The favourable soil moisture availability under this treatment favoured higher NPK uptake, which in turn reflected on increased plant height and LAI. The increased LAI helped in efficient photosynthesis and their translocation to the various parts of the plant when consequently increased the DMP. This was in close

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confirmity with the findings of Patil and Sheelavantar (2004) and Dwyer and Stewart (1985). Bharathi *et al.* (1997) also reported maximum DMP with increased soil moisture availability and reduction in DMP under moisture stress conditions. This might be attributed to the reduced velocity of runoff from broad bed and furrow due to uniform slope, which might have resulted in higher opportunity time for infiltration in broad bed and furrow on a vertisols than other methods (Acharya and Hati, 2002) (Table 3).

The influence of stress management practices on rainfed cotton revealed that at 30 and 60 DAS, soil application of pusa hydrogel @ 5 kg ha^{-1} treated plots (S_1, S_2, S_3, S_4 and S_5) registered higher DMP than the untreated control plot (S_6). Further, the significantly higher DMP of 3778, 4957 (2016) and 4153, 5500 kg ha^{-1} (2017) were recorded under soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} (S_4) at 90 and 120 DAS, respectively. The increase in dry matter production was due to increase in carbohydrates, proteins, total

Table 3 : Effect of *in-situ* moisture conservation and stress management practices on dry matter production (kg ha^1) of rainfed cotton during *rabi* 2016 and 2017

Treatment	*Rabi* 2016				*Rabi* 2017			
	I_1	I_2	I_3	Mean	I_1	I_2	I_3	Mean
S_1	4,515	4,190	3,885	4,197	5,111	4,646	4,180	4,646
S_2	4,835	4,528	4,240	4,534	5,616	5,313	4,992	5,307
S_3	4,676	4,391	4,126	4,398	5,406	5,149	4,896	5,150
S_4	5,173	4,950	4,748	4,957	5,848	5,500	5,153	5,500
S_5	4,612	4,272	3,946	4,277	5,402	4,923	4,386	4,904
S_6	4,289	3,892	3,507	3,896	4,734	4,372	4,010	4,372
Mean	4,683	4,371	4,075		5,353	4,984	4,603	
	I	**S**	**I at S**	**S at I**	**I**	**S**	**I at S**	**S at I**
S.Ed	108	82	133	115	103	89	156	126
CD (P < 0.05)	304	168	377	238	309	184	473	260

amino acids and other biochemical and physiological parameters especially in presence of hydrogel polymer. Similarly, an increase in dry matter production due to hydrogel polymer was reported by Akhter *et al.* (2004) in barley and Yazdani *et al.* (2007) in soybean. Hydrogel sticks to the roots of the plants and when the soil moisture falls as the temperature rises, the gel sheds water to nourish the crop. The stored water in the polymer was released as required by the crop and used to meet the crop water requirement and also resulted with better aggregation of soil particles, improved water use efficiency, reduced infiltration rate, increased water holding capacity and decreased evaporation losses. Super absorbent polymer can increase soil's water holding capacity and delaying water stress in plants and providing a buffer against product loss during the time of stress condition (Mohammad and Fardin, 2012). Foliar application of PPFM enhanced the production of growth promoting substances *viz.,* cytokinin, IAA and GA. The combined effect of auxin and cytokinin, which maintained a balanced growth of shoot and root system (Suresh Reddy, 2002). The contribution of the IAA, cytokinin and GA to the enhanced root growth enabling the plants to absorb more nutrients ultimately resulting in better growth and dry matter production. The growth promoting substances and methanol releases by PPFM especially through apical plant tissues. This is in agreement with the findings of Radha *et al.* (2009). The increased DMP through application of PPFM could also be attributes by the role of methanol as a carbon source to increase carboxylase reaction and enhanced photosynthetic rate (Kumar *et al.*, 1999; Zbiec *et al.*, 2003). Under interaction, the maximum DMP (32 and 31% over I_3S_6, respectively during 2016 and 2017) were recorded under the combination of broad bed and furrows with soil application of pusa hydrogel @ 5 kg ha^{-1} + PPFM @ 500 ml ha^{-1} (I_1S_4) at 90 and 120 DAS. This was due to the cumulative effect of higher leaf area index under this treatment coupled with favourable soil moisture and microclimate due to broad bed and furrows with pusa gel.

Effect of *in-situ* moisture conservation and stress management practices on yield attributes of cotton

Number of sympodia plant^{-1}

The number of sympodial branches per plant was found significantly higher with BBF (10.5 in 2016 and 13.0 in 2017) than other treatments. Due to increased and continuous availability of soil moisture under broad bed and furrows system, the growth was enhanced and consequently the number of sympodial branches increased. Regarding moisture stress management practices, soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} recorded significantly higher number of sympodial

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branches plant^{-1} (11.0 in 2016 and 13.2 in 2017). The application of super absorbent polymer significantly increased number of leaves, root biomass and number of sympodial branches over control. These results confirmed the earlier findings of (Bai *et al.*, 2010). Further, due to PPFM foliar application, the combined effect of auxin and cytokinin maintained a balanced growth of shoot and root system. When roots become more extensive due to the action of auxins, subsequently, cytokinins of the plant act as signals in the shoot system to produce more branches as reflected by the higher number of sympodial branches under this treatment. This is in agreement with the findings of Abd El-Gawad *et al.* (2015) (Table 4).

Table 4 : Effect of *in-situ* moisture conservation and stress management practices on sympodia plant^{-1} of rainfed cotton during *rabi* 2016 and 2017

Treatment	*Rabi* 2016				*Rabi* 2017			
	I_1	I_2	I_3	Mean	I_1	I_2	I_3	Mean
S_1	9.8	9.1	8.4	9.1	12.4	11.2	10.3	11.3
S_2	10.7	10.1	9.5	10.1	13.6	12.4	11.3	12.4
S_3	10.4	9.7	9.4	9.8	13.5	12.1	10.9	12.2
S_4	12.5	10.9	9.7	11.0	14.9	13.2	11.5	13.2
S_5	10.5	9.4	8.6	9.5	12.9	12.0	11.2	12.0
S_6	9.0	8.2	7.5	8.2	10.6	10.2	9.9	10.2
Mean	10.5	9.6	8.9		13.0	11.9	10.9	
	I	**S**	**I at S**	**S at I**	**I**	**S**	**I at S**	**S at I**
S.Ed	0.2	0.2	0.3	0.3	0.2	0.3	0.5	0.5
CD (P < 0.05)	0.8	0.4	0.9	0.6	0.5	0.6	1.0	1.0

management practices, soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} recorded 32 and 40% increased bolls plant^{-1} during 2016 and 2017 respectively over control**.** Adequate supply of assimilates at the reproductive stage was one among the reasons for increased number of bolls per plant. In addition, PPFM spray enhanced the cytokinin synthesis and resulted in delayed senescence and maintenance of cell division after boll formation. The positive influence of PPFM spray in increasing the number of bolls per plant also recorded by Madhaiyan *et al.* (2006b) (Table 5).

Table 5 : Effect of *in-situ* moisture conservation and stress management practices on bolls plant^{-1} of rainfed cotton during *rabi* 2016 and 2017

Treatment	*Rabi* 2016				*Rabi* 2017			
	I_1	I_2	I_3	Mean	I_1	I_2	I_3	Mean
S_1	13.7	11.8	9.7	11.7	16.1	13.7	11.2	13.7
S_2	17.7	15.3	13.7	15.6	20.9	19.0	17.2	19.0
S_3	15.4	13.1	10.7	13.1	17.5	16.7	15.8	16.7
S_4	20.0	18.2	15.3	17.8	23.8	20.8	17.7	20.8
S_5	16.8	13.8	10.7	13.8	18.3	15.2	12.0	15.2
S_6	11.1	8.8	6.4	8.8	11.9	9.1	7.3	9.4
Mean	15.8	13.5	11.1		18.1	15.8	13.5	
	I	**S**	**I at S**	**S at I**	**I**	**S**	**I at S**	**S at I**
S.Ed	1.0	1.0	1.6	1.4	0.6	0.6	1.1	1.0
CD (P < 0.05)	2.2	2.0	4.7	2.9	1.6	1.2	3.7	2.1

Number of bolls plant^{-1}

Boll production and their retention in the plant are controlled by soil moisture conditions. Early energy translocation to the reproductive parts under higher moisture supply may be the reason for increased boll production under BBF. Hence, the higher number of bolls plant^{-1} (15.8 in 2016 and 18.1 in 2017) was noticed under broad bed and furrows than the other moisture conservation methods during both the years of experimentation. This might also be ascribed to higher retention of moisture in boll development stage and better aeration in rooting zone. This leads to more numbers of bolls productions. Similar results were also reported by Kuotsu *et al.* (2014). Among the stress

Mean boll weight (g boll^{-1})

The adoption of BBF method recorded significantly higher boll weight of 3.64 (2016) and 3.76 g boll^{-1} (2017) compared to other *in-situ* moisture conserving land management practices. This may be due to increased DMP and better translocation of photosynthates from source to sink and resulted in better development of bolls. Regarding the stress management practices, soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} recorded 13-15% higher boll weight than the control. This might be due to application of super absorbent polymers helped to alleviate drought stress through retention of more moisture

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in the rooting zone and increased number and weight of bolls. Similar finding were also reported by Allah Dadi *et al.* (2005). Khadem *et al.* (2010) also reported that the addition of super absorbent polymer linearly increased the seed weight of corn and soybean crops. Further, foliar application of PPFM, increased total dry matter production of cotton due to better translocation of photosynthates and led to better boll development and boll weight. These results are in conformity with observations made by Madhaiyan *et al.* (2006b) (Table 6).

Table 6 : Effect of *in-situ* moisture conservation and stress management practices on boll weight (g boll^{-1}) of rainfed cotton during *rabi* 2016 and 2017

Treatment	*Rabi* 2016				*Rabi* 2017			
	I_1	I_2	I_3	Mean	I_1	I_2	I_3	Mean
S_1	3.47	3.30	3.28	3.35	3.63	3.39	3.25	3.42
S_2	3.88	3.60	3.44	3.64	3.96	3.70	3.47	3.71
S_3	3.59	3.42	3.31	3.44	3.77	3.47	3.27	3.50
S_4	3.99	3.74	3.51	3.75	4.07	3.84	3.63	3.85
S_5	3.51	3.39	3.31	3.40	3.65	3.47	3.33	3.48
S_6	3.37	3.21	3.19	3.26	3.46	3.28	3.20	3.31
Mean	3.64	3.44	3.34		3.76	3.53	3.36	
	I	**S**	**I at S**	**S at I**	**I**	**S**	**I at S**	**S at I**
S.Ed	0.08	0.11	0.19	0.19	0.10	0.11	0.20	0.18
CD (P < 0.05)	0.20	0.13	NS	NS	0.24	0.16	NS	NS

Effect of *in-situ* moisture conservation and stress management practices on seed cotton yield

Yield is contributed by different yield components and any influence made by extraneous factor, will alter the yield significantly. In the present study, increased seed cotton yield could be attributed to better crop growth and yield components due to consistent soil moisture availability due to combined influence of BBF, soil conditioner and foliar application of PPFM. The broad bed furrow system significantly influenced the seed cotton yield as compared to other land configuration. BBF recorded significantly higher seed cotton yield of 1,246 (2016) and 1,590 kg ha^{-1} (2017) which was 23% (2016) and 19% (2017) higher as compared to compartmental bunding (Table 2). Increment in seed cotton yield was due to more soil moisture availability at the root zone which favoured better crop growth, more nutrient uptake and higher translocation leading to production of larger leaf area index which was responsible for harvesting more solar energy. This coupled with higher stomatal conductance and transpiration rate resulted in accumulation of more dry matter and yield components and ultimately the seed cotton yield. This is in similarity with the findings of Muralidaran and Solaimalai (2005).

Higher seed cotton yield was realized with complementary alliance of *in-situ* moisture conservation measures with stress management practices in the present study. Significant influence by stress management practices also recorded with soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1}, which registered higher seed cotton yield of 1,394 and 1,786 kg ha^{-1} during 2016 and 2017, respectively.

Table 7 : Effect of *in-situ* moisture conservation and stress management practices on seed cotton yield (kg ha^{-1}) under rainfed condition during *rabi* 2016 and 2017

Treatment	*Rabi* 2016				*Rabi* 2017			
	I_1	I_2	I_3	Mean	I_1	I_2	I_3	Mean
S_1	1,112	1,039	979	1,043	1,457	1,373	1,299	1,376
S_2	1,369	1,260	1,084	1,238	1,686	1,579	1,475	1,580
S_3	1,229	1,160	1,065	1,151	1,582	1,484	1,389	1,485
S_4	1,580	1,350	1,253	1,394	1,943	1,785	1,631	1,786
S_5	1,189	1,094	992	1,092	1,544	1,472	1,411	1,476
S_6	998	884	724	869	1,328	1,107	893	1,109
Mean	1,246	1,131	1,016		1,590	1,467	1,350	
	I	**S**	**I at S**	**S at I**	**I**	**S**	**I at S**	**S at I**
S.Ed	33	33	61	57	32	48	71	68
CD (P=0.05)	90	67	138	116	117	100	178	141

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The increased seed cotton yield under soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} was 60% during 2016 and 61% in 2017 over control. This may be due to increase in growth and yield related attributes. Further, it could be because of sufficient availability of soil moisture due to super absorbent polymer under water stress condition, which in turn lead to better translocation of water, nutrients and photo assimilates and finally better plant development. Similar findings were also reported by Sivapalan (2001) and El-Hady *et al.* (1981) under water stress condition. However during stress, presence of photosynthetically active leaf area due to moisture availability by SAP exhibited more chlorophyll meter readings was also positively correlated to yield. Similar findings were reported by Johnkutty and Palaniappan (1995).

The present study revealed that foliar application of PPFM when combined with soil application of pusa hydrogel resulted in enhanced seed cotton yield. The enhancement of yield was associated with increased number of bolls per plant through improved boll retention. Significant increase in seed cotton yield with the application of PPFM was reported by Dhale *et al.* (2010). The beneficial influence by PPFM may be attributed to several factors such as release of growth promoting substances like auxins, particularly indole-3-acetic acid (IAA) and indole-3-pyruvic acid, zeatin, zeatinriboside, proliferation of beneficial organisms in the phyllosphere and reacted cytokinins by methylotrophs were enhance the plant growth of crops due to vegetative growth of the plant attributed to increase in the yield of a crop. Hence, it could be concluded that foliar application of PPFM favourably influenced the seed cotton yield.

The combination of BBF with soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} recorded higher yield with an increase of 133 and 150% during 2016 and 2017, respectively than the compartmental bunding with control. This may be due to the favourable environment provided by the *in-situ* moisture conservation and stress management practices, which enhanced the growth and yield attributing characters of cotton during both the years of study.

Conclusion

Broad bed and furrows combined with foliar application of PPFM spray at 500 ml ha^{-1} resulted in significantly taller plants, higher dry matter production, LAI, yield attributing characters like number of sympodia plant^{-1}, number of bolls plant^{-1}, boll weight and seed cotton yield under rainfed areas. The higher values indicate that the moisture conservation and stress management practices improved growth rate performance leading to higher yield potential of rainfed cotton. Broad bed and furrows combined with soil application of pusa hydrogel @ 5 kg ha^{-1} + foliar spray of PPFM @ 500 ml ha^{-1} during the stress period was found to be the best agronomic management practice in order to enhance yield in cotton under rainfed vertisols.

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